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PPO-Based Control of an Active Motion Surface for Parcel Singulation

Final policy selection and robustness assessment

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This presentation summarizes our final work on PPO-based control on an Active Motion Surface for parcel singulation

Contents

1. Environment, reward, and physical KPI

AMS model, downstream reward, and edge-based metric

2. PPO baseline

2000-episode reference configuration and KPIs

3. Bayesian-optimized PPO Policy

High-fidelity BO and best candidate selection

4. Robustness assessment

Multiseed validation and non-nominal stress campaign

5. Conclusions and Future improvements

As shown in the pipeline, the focus is not only on training a policy, but also on selecting a final PPO configuration and validating it through physical metrics, multi-seed analysis, and non-nominal stress tests

PPO acts on the AMS; singulation is evaluated downstream

Implemented system

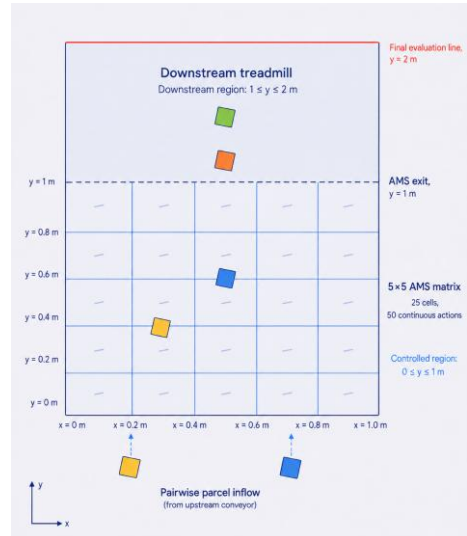
- 5×5 AMS matrix with 0.2 m cell side
- 25 cells × 2 commands = 50 continuous actions components
- Commands: actuator rotation and velocity magnitude

Nominal parcel flow

- 10 parcels per episode, generated in pairs
- Pair generation period: 0.75 s
- Downstream treadmill speed: 1.9 m/s

Physical regions

- PPO-controlled AMS region: $0 \leq y \leq 1$ m
- Passive downstream region: $1 \leq y \leq 2$ m
- Final evaluation line: $y = 2$ m



The system consists of a 5 by 5 Active Motion Surface followed by a treadmill.

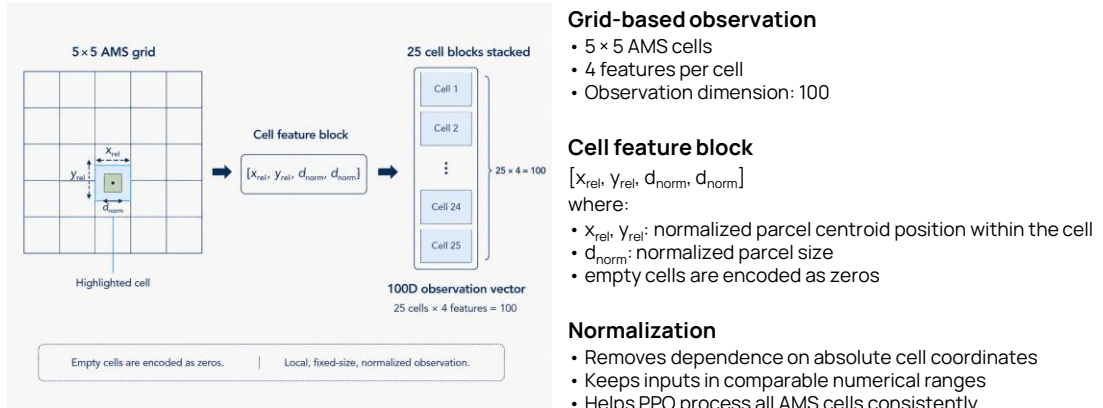
Each cell of AMS has two commands, actuator rotation and velocity magnitude, giving 50 continuous action components.

In the nominal case 10 parcels are generated in pairs every 0.75 seconds and the downstream treadmill moves at 1.9 meters per second.

The PPO agent acts only on the AMS region, between zero and one meter in the longitudinal direction.

Singulation is instead evaluated downstream, at the final line $y=2$ meters.

The PPO agent observes a normalized local representation of the AMS



The observation is a compact grid-based representation of the AMS.

For each of the 25 cells, four features are stored: the normalized local parcel position and the normalized parcel size.

Empty cells are encoded as zeros, so the total observation dimension is 100.

The important point is that the observation is local and normalized.

This avoids using absolute cell coordinates directly and helps PPO process all cells in a consistent numerical range.

The training reward^[*] promotes downstream pairwise separation

Rewarding logic

$$1 \leq y \leq 2 \text{ m}$$

At each step:

1. Select parcels in the downstream region
2. Sort them by longitudinal position
3. Compute edge gaps between consecutive parcels
4. Convert each gap into a local score
5. Use the worst pair as instantaneous reward

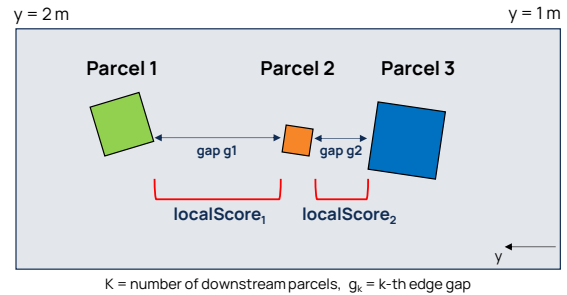
Bonus:

+20 if all gaps are positive and at least three parcels are present.

Mathematical expression

$$r = \min_k(\text{localScore}_k) + \text{bonus}$$

$$\text{localScore}_k = \min(10 \cdot K \cdot \text{atan}(g_k) + K, 6)$$



[*]: Roveda, Maccarini, Pura, Reiso & Spahiu (2025) Optimizing Parcels Sorting Through Reinforcement Learning for Intralogistics.

The training reward is computed in the downstream region, between the AMS exit and the final evaluation line.

At each step, parcels in this region are sorted longitudinally, and edge gaps are computed between consecutive parcels.

These gaps are converted into local scores, and the instantaneous reward is governed by worst separated pair.

A bonus is added only when all observed gaps are positive and at least three parcels are present.

This makes the reward physically related to singulation, but it is still only the learning signal used by PPO.

Final physical performance is evaluated using parcel-edge gaps

Evaluation line

- Final parcel configuration at $y = 2$ m
- After the downstream region

Edge-based pair gap

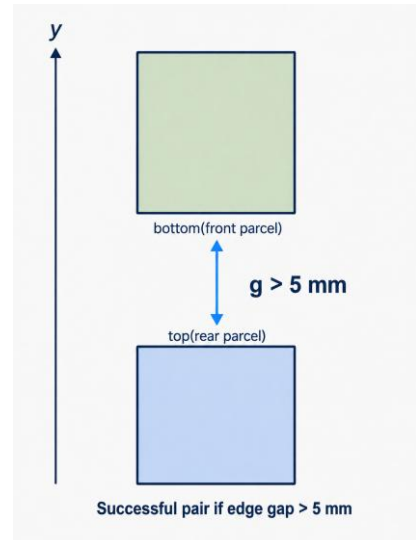
- $g = \text{bottom}(\text{front parcel}) - \text{top}(\text{rear parcel})$
- Centroid distance is not used

Success criterion

- Successful pair: $g > 5$ mm
- Reported metric: percentage of successful consecutive pairs



The reward guides learning;
final edge gaps evaluate physical singulation.



For the final evaluation, we do not use the Average Reward, but the main physical metric, which is the edge-based final accuracy.

At $y=2$ m, two consecutive parcels are considered successfully separated only if the edge gap is larger than 5 millimeters.

This is stricter than the centroid-distance criterion, but avoids overlapping and tangent edges.

The distinction between the reward and the physical KPI is central to our project.

The 2000-episode PPO baseline was effective but still improvable

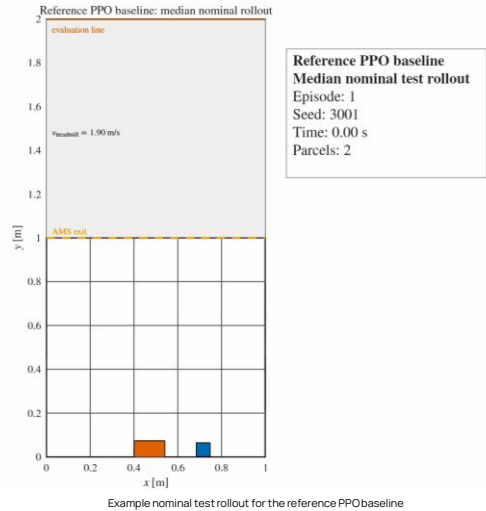
Reference PPO setup

Experience horizon: 500
 Clip factor: 0.20
 Actor/Critic LR: $1e-4$
 GAE factor: 0.95
 Discount factor: 0.99
 Entropy weight: 0.01
 Training seed: 1

Reference PPO physical performance

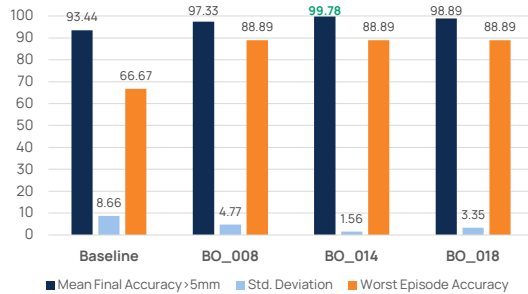
Mean final accuracy >5mm: 93.44 %
 Std. Deviation: 8.66 pp
 Worst episode accuracy: 66.67 % → **Margin to improve**
 Mean Throughput: 8250.1 pph
 Mean max overlap: 0.0500m → **Margin to improve**
 Mean Collision count: 20.53
 Evaluation seed base: 3000

High average performance was achieved, but worst-case episodes and residual overlap motivated **PPO tuning**.



- **Steps:**
 - Validate environment, reward, evaluation metrics
 - Train PPO baseline
 - BO for hyperparameters tuning
 - Multiseed validation and stress testing
- **Baseline PPO performance:**
 - Meaningful singulation (high accuracy)
 - High standard deviation
 - Low worst episode accuracy
 - High mean max overlap (max(min(pair-wise gaps))
 - BO to improve performance

Bayesian Optimization candidates outperformed the reference baseline



PPO parameter

Experience horizon:
Clip factor:
Actor LR:
Critic LR:
GAE factor:
Discount factor:
Entropy weight:
Epochs:

Baseline

500
0.20
1e-4
1e-4
0.95
0.99
0.01
3

BO_014

400
0.098
5,66e-5
2,96e-4
0.959
0.995
1,60e-4
4

Parameter shift interpretation

- Smaller **clip factor** and lower **actor LR** suggest more conservative policy updates.
- Higher **critic LR** and one extra **epoch** improve value-function fitting during PPO updates.
- Much lower **entropy weight** reduces exploration once the policy has learned a stable singulation behavior.
- Slightly higher **discount factor** keeps downstream separation more relevant in the optimization.

Set	Mean overlap [m]	Throughput [pph]	Collisions
Baseline	0,0500	8250,1	20,53
BO_008	0,0222	8052,1	5,05
BO_014	0,00133	7874,2	1,00
BO_018	0,00352	8244,7	0,25

Slide 1

Θ: BO search space

θ: PPO hyperparameters

- Mean final accuracy > 5 mm $\rightarrow N(g_i > 5\text{mm})/N_{\text{pairs}} * 100$
 - $g_i \rightarrow y_{\text{bottom,front}} - y_{\text{top,rear}}$
- Environment, observation, reward, physical KPI, network architecture fixed
- Goal: maximise mean final edge gap accuracy > 5mm, not average reward
- Candidate BO_14 achieved
 - Highest mean final accuracy
 - Lowest std
 - Lowest overlap depth

Slide 2

- Lower clip factor and actor LR: more conservative actor policy updates
- Higher critic LR: the critic is trained more aggressively, providing a better estimate of the expected future reward used to update the actor.
- Lower entropy weight: reduced exploration once the stable singulation policy has been learned
- Higher discount factor: helps PPO account for separation effects that appear after parcels leave the AMS region.

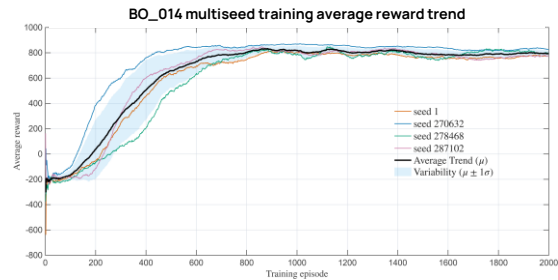
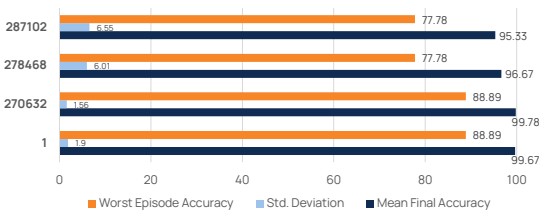
Multiseed validation confirmed the robustness of BO_014

Multiseed validation

- Final configuration: BO_014
- 4 independent PPO training seeds
- 100 nominal evaluation episodes per seed

Training diagnostic

- Reward trends indicate stable PPO training across seeds.
- AverageReward is not used as the final success metric.



Physical robustness

- Mean accuracy across seeds: 97.86%
- Std across training seeds: 2.22 pp
- Worst episode across all seeds: 77.78%

BO_014 is not dependent on a single PPO training realization and remains **physically effective** across different random seeds.

- Same configuration trained with 4 independent seeds and evaluated on 100 episodes
- Reward indicates stable learning but not used as final success metric → physical metrics are used
- Results
 - Mean 97,8% accuract accuracy across seeds
 - 2.22 percentage points std
 - Worst episode accuracy: 77.78%

Non-nominal stress tests evaluated the operating limits of BO_014

Evaluation-only stress campaign

- Final configuration: BO_014
- 4 trained agents from the multiseed analysis
- 4 agents × 100 episodes per scenario
- Evaluation seed base: 2044876
- No retraining
- No reward, no PPO changes



Treadmill
velocity



Generation
frequency



Parcel
shape



Total parcels
per episode



Parcels per
generation

The goal is to distinguish **robust** generalization from actual **operating limitations**.

The next step was to evaluate the operating limits of the 14th candidate under non-nominal conditions.

This campaign was evaluation-only:

four trained agents from the multiseed analysis were tested, with 100 episodes per scenario, without further retraining or reward/PPO settings changing.

The stress categories were:

- treadmill velocity,
- generation frequency,
- parcel shape,
- total number of parcels per episode,
- and number of parcels generated simultaneously.



Reduced treadmill velocity caused moderate degradation

Nominal case

Mean Final Accuracy > 5mm: 98,28%
 Worst Agent Accuracy: 96,33%
 Mean Max Overlap: 9,9 mm
 Throughput: 8027 pph

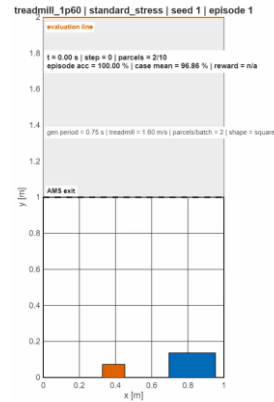
Treadmill-1p60

Mean Final Accuracy > 5mm: 96,86% **-1,42 pp**
 Worst Agent Accuracy: 93,67% **-2,66 pp**
 Mean Max Overlap: 14,1 mm **+4,2 mm**
 Throughput: 7874 pph **-153 pph**

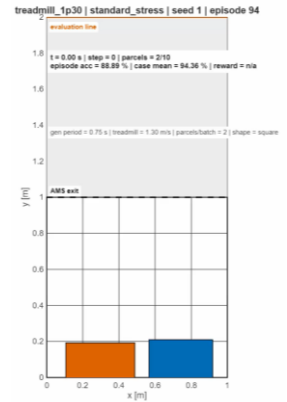
Treadmill-1p30

Mean Final Accuracy > 5mm: 94,36% **-3,92 pp**
 Worst Agent Accuracy: 88,00% **-8,33 pp**
 Mean Max Overlap: 24,1 mm **+14,2 mm**
 Throughput: 7662 pph **-365 pph**

Treadmill velocity = 1,60 m/s



Treadmill velocity = 1,30 m/s



Reduced downstream speed degrades separation quality, but it is not the most critical operating limitation.

Reducing the downstream treadmill velocity caused a moderate degradation. At 1.60 m/s, the mean accuracy decreased by 1.42 percentage points, while at 1.30 m/s the decrease was 3.92 percentage points. The overlap depth increased, but the policy still maintained a reasonably high accuracy.

Therefore, downstream speed reduction affects separation quality, but it does not represent the most critical operating limitation.



Higher generation frequency revealed a temporal inflow limit

Nominal case

Mean Final Accuracy > 5mm: 98,28%
 Worst Agent Accuracy: 96,33%
 Mean Max Overlap: 9,9 mm
 Throughput: 8027 pph

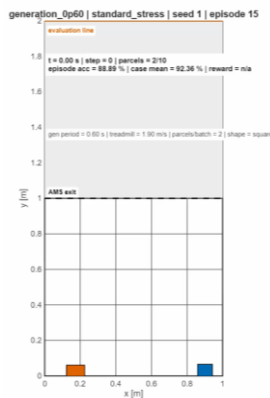
Generation-0p60

Mean Final Accuracy > 5mm: 92,36% **-5,92 pp**
 Worst Agent Accuracy: 90,89% **-5,44 pp**
 Mean Max Overlap: 57,9 mm **+48,0 mm**
 Throughput: 9243 pph **+1216 pph**

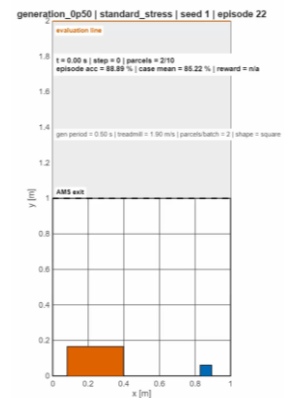
Generation-0p50

Mean Final Accuracy > 5mm: 85,22% **-13,06 pp**
 Worst Agent Accuracy: 77,67% **-18,66 pp**
 Mean Max Overlap: 112,9 mm **+103,0 mm**
 Throughput: 10360 pph **+2333 pph**

Generation frequency = 0,60 s



Generation frequency = 0,50 s



The policy degrades when parcels enter the AMS **too frequently**, leaving less time to **recover longitudinal separation**.

The generation-frequency cases are more critical.

In fact, reducing the generation period from the nominal 0.75 s to 0.60 s already decreased the mean accuracy by 5.92 percentage points.

At 0.50 seconds, the decrease reached 13.06 percentage points, and the mean overlap depth increased by more than 10 centimeters.

This shows a temporal inflow limit: when parcels enter the AMS too frequently, the policy has less time to recover longitudinal separation.



BO_014 generalized well to rectangular parcel shapes

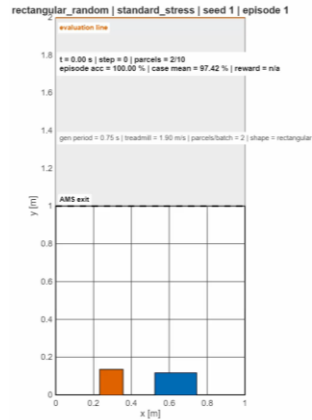
Nominal case

Mean Final Accuracy > 5mm: 98,28%
 Worst Agent Accuracy: 96,33%
 Mean Max Overlap: 9,9 mm
 Throughput: 8027 pph

Rectangular parcels

Mean Final Accuracy > 5mm: 97,42% **-0,86 pp**
 Worst Agent Accuracy: 95,00% **-1,33 pp**
 Mean Max Overlap: 17,9 mm **+8,0 mm**
 Throughput: 8036 pph **+9,0 pph**

Rectangular dimensions randomly sampled within the nominal parcel-size bounds: **[0.05, 0.40] m**



The policy is **not strongly tied** to the nominal **square-parcel** assumption.

The decision to test a rectangular-parcel case was taken to understand whether the policy was too dependent on the nominal square shape assumption.

The actual degradation was limited, as the mean accuracy decreased by only 0.86 percentage points, and the worst-agent accuracy remained at 95 percent.

This suggests that the selected policy generalizes well to moderate shape variations, at least within the same parcel-size bounds used for the nominal configuration.



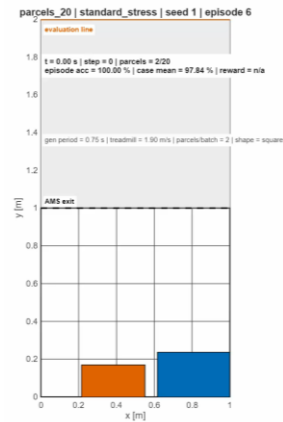
Increasing the total number of parcels did not collapse performance

Nominal case

Mean Final Accuracy > 5mm: 98,28%
 Worst Agent Accuracy: 96,33%
 Mean Max Overlap: 9,9 mm
 Throughput: 8027 pph

Parcels_20

Mean Final Accuracy > 5mm: 97,84% **-0,44 pp**
 Worst Agent Accuracy: 95,84% **-0,49 pp**
 Mean Max Overlap: 25,1 mm **+15,2 mm**
 Throughput: 8738 pph **+711 pph**



The policy **scales well** with the total number of parcels when the nominal pairwise generation structure is preserved.

When the total number of parcels was increased to 20, the policy did not collapse. The mean accuracy decreased by only 0.44 percentage points, and the worst-agent accuracy remained close to nominal. The overlap depth increased, but the result shows that the policy scales well with the total number of parcels when the nominal pairwise generation structure is preserved.



Three-parcel generation was the most severe stress condition

Nominal case

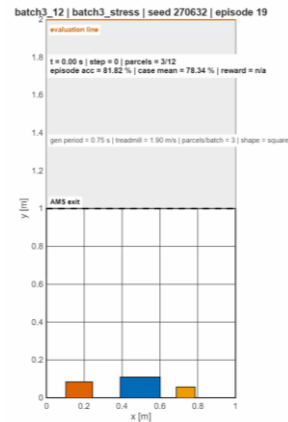
Mean Final Accuracy >5mm: 98,28%
 Worst Agent Accuracy: 96,33%
 Mean Max Overlap: 9,9 mm
 Throughput: 8027 pph

Batch3_12

Mean Final Accuracy >5mm: 78,34% **-19,94 pp**
 Worst Agent Accuracy: 72,00% **-24,33 pp**
 Mean Max Overlap: 112,8 mm **+102,9 mm**
 Throughput: 11481 pph **+3454 pph**

At each generation event, 3 parcels enter the system in a **left-center-right configuration**. To avoid artificial initial overlaps and preserve AMS-grid compatibility, the upper parcel-size bound was reduced to **0.30 m**. The number of parcels per episode was set to 12, corresponding to **4 complete batches**.

The main difficulty is not only temporal **inflow rate**, but also **simultaneous spatial crowding** at generation.



The most severe stress condition was the three-parcel generation case.

Here, each generation event produced three parcels in a left-center-right configuration.

To avoid artificial initial overlaps and preserve AMS-grid compatibility, the upper parcel-size bound was reduced to 0.30 m, and the episode contained 12 parcels, corresponding to four complete batches.

This case reduced the mean accuracy by almost 20 percentage points, and the worst-agent accuracy by more than 24 percentage points.

The key message is that the difficulty is not only the temporal inflow rate, but also simultaneous spatial crowding at generation.

Final outcome: strong nominal policy, clear robustness target



Validated formulation

- Environment, reward, and physical KPIs were verified before final training.
- AverageReward was used only as a training diagnostic.



Selected policy

- BO improved PPO over the reference baseline.
- BO_014 achieved strong nominal and multiseed performance.



Observed limitation

- Shape and parcel-count variations were handled well.
- Dense and simultaneous inflow remained the main robustness limit.

Next steps

Mixed high-density training · Crowding-aware observation · Congestion-aware reward terms · Curriculum learning

Future work should target robustness to **dense inflow**, where the current policy shows its clearest **operating limit**.

To conclude:

- The environment, reward, and physical KPIs were verified before final training, and AverageReward was used only as a training diagnostic.
- Bayesian Optimization improved PPO over the reference baseline, and BO_014 achieved strong nominal performance and robust multiseed behavior.
- The stress campaign showed that shape variations and a higher total number of parcels are handled well. The clearest limitation appears under dense and simultaneous inflow.

Therefore, future work should focus on mixed high-density training, crowding-aware observations, congestion-aware reward terms, and possibly curriculum learning.

So, we can agree that the key takeaway is that BO_014 is a strong nominal policy, but improving robustness to dense inflow is the key direction for further development.